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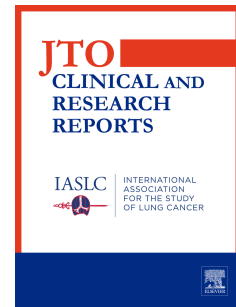
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Abstract**Background:**

Local treatment options for progressive lung cancer or metastases located in central or middle thirds of the lungs are limited, especially if progression occurs following radiation. In these patients, local thermal ablative therapies are fraught with complications. We investigated the safety and efficacy for local control in progressive lung and metastatic cancer by bronchoscopic pulsed electric field (PEF) ablation.

Methods:

We performed analysis of prospectively collected data of all the patients with lung and metastatic cancer to the lungs treated by bronchoscopic PEF ablation at our institution between October 2024 to April 2025.

Results:

Seven lesions in six consecutive patients were treated by PEF ablation using robotic bronchoscopy with three-dimensional imaging. There were three patients with non-small cell lung cancer (NSCLC), and three with colorectal cancer, urothelial cancer, and thyroid cancer. All patients had progressed on standard oncologic therapies and five had received radiation to the lesions. The median tumor size was 2.1 cm (range, 1.1-4 cm) and all were located in the central or middle-thirds of the lungs. At the median follow up of 9.7 (5.4-10.1) months, four (57.1%) lesions increased in size, two (28.5%) decreased, and one (14.3%) resolved. The decreased size or resolution of lesions was seen in the three patients with NSCLC. No direct complications due to PEF were encountered, but one patient required prolonged mechanical ventilation related to anesthesia.

Conclusions:

In patients with lung cancer or lung metastases, bronchoscopic PEF ablation is safe and associated with mixed local control.

Keywords:

Lung cancer, bronchoscopy, pulsed electric field ablation

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Abbreviations:

¹⁸F-fluorodeoxyglucose: FDG

Electrocardiogram: EKG

Interquartile range: IQR

Microwave: MWA

Non-small cell lung cancer: NSCLC

PEF: Pulsed electric field

Positron emission tomography: PET

Radiofrequency ablation: RFA

Response Evaluation Criterion in Solid Tumors: RECIST

SBRT: Stereotactic body radiation therapy

Three-dimensional: 3D

Introduction

Lung cancer is the leading cause of cancer-related mortality and other primary malignancies with lung metastases have a poor prognosis.^{1,2} Despite recent oncologic advances, including immunotherapy, tyrosine kinase inhibitors, and stereotactic body radiation therapy (SBRT), the outcomes for patients with progressive lung cancer and metastases remain poor.³⁻⁵ In patients with progressive local lung cancer or oligometastatic cancer in lungs, local thermal ablation therapies, including radiofrequency ablation (RFA), microwave (MWA), and cryoablation, have been used with varying success and safety.^{6,7} The lesions in the central and middle-thirds of the lungs can be challenging to access percutaneously and central blood vessels increase the risk of excessive bleeding.⁸ Bronchoscopy, especially with robotic navigation platforms and three-dimensional (3D) imaging, may provide a safe and effective access to these lesions.⁹ But the experience with bronchoscopic thermal ablation, including RFA and MWA, has been fraught with multiple complications, including life-threatening hemorrhage and death.^{7,10,11} Pulsed electric field (PEF) ablation offers a safer, non-thermal alternative in which high-voltage electrical pulses are passed through a lesion, leading to alteration in cell transmembrane potential and cell apoptosis.^{12,13} In addition, it releases non-denatured proteins into the tumor microenvironment which may enhance a native immunogenic response.¹⁴⁻¹⁶ Because the cell death occurs slowly and the extracellular matrix is preserved, the risk of complications is lower when used in close proximity to critical structures such as blood vessels, bronchioles, and pleura as compared to other locally delivered ablative therapies.^{17,18} The Federal Drug Administration (FDA) cleared a flexible needle (INUMI Flex endoscopic needle, Galvanize Therapeutics Inc, Redwood City, CA) that can be used with Aliya electrosurgical system to perform bronchoscopic PEF ablation of lung lesions.¹⁹ To-date, the short-term safety of bronchoscopic PEF has been favorable as described in small studies; however, there has been no report on the efficacy for local control of malignancy.¹⁷⁻²⁰ The objective of this study is to describe the long-term clinical safety and early efficacy data for local control by bronchoscopic PEF ablation in patients with progressive non-small cell lung cancer (NSCLC) and metastatic cancer to the lungs.

Methods

Study Design and Subjects:

We prospectively collected data on consecutive patients treated clinically with bronchoscopic PEF at our institution between October 2024 to April 2025. All the patients had lung cancer or metastatic cancer to lungs from other primaries, treated previously with radiation and/or systemic therapy, and had experienced radiographic cancer progression. To be eligible for PEF ablation, these patients were discussed by their multidisciplinary clinical team and referred for evaluation by interventional pulmonology. All the patients provided consent for the clinical procedure. The study was approved by the institutional IRB (Pro00116887) and informed consent for enrolment in the research project was waived. The safety and local efficacy data were collected along with other relevant information. The short-term and long-term safety were defined as adverse events or complications within and beyond 30 days, respectively.

Bronchoscopy

All patients underwent bronchoscopy using shape-sensing robotic platform (Ion, Intuitive Surgical, Sunnyvale, CA) with 3D imaging (Cios Spin, Siemens Medical Solutions, Malvern, PA) for tool-in-lesion confirmation. Patients underwent general anesthesia with endotracheal intubation. A flexible bronchoscope was first introduced in the airway for inspection. It was followed by insertion of robotic bronchoscope catheter with navigation to the target lesion using 3D imaging assistance. A diagnostic sample was obtained with a flexible needle, reviewed by rapid-on-site evaluation (ROSE), and malignancy was confirmed before ablation was performed.

PEF Ablation

For ablation of the lung lesions, Aliya PEF System (Galvanize Therapeutics, Inc, Redwood City, CA) was used with an INUMI Flex Needle which delivers short duration, high voltage (3kV), biphasic electrical pulses. INUMI Flex needle is 160 cm long flexible catheter with a distal adjustable 21-gauge, partially-

insulated needle, 3.1 cm in total length with 0.5 cm active distal tip. Per instructions for use for INUMI Flex needle based on preclinical studies, the volume of ablation zone is approximately $1.4 \pm 0.2 \times 1.1 \pm 0.1 \times 1.1 \pm 0.1 \text{ cm}^3$ or 0.88 cm^3 . Once the needle is in target, the stylet of the catheter is removed. An electrode filament is inserted into the proximal catheter and attached to the generator. The generator is connected to a cardiac monitor which is attached to the patient by standard four chest leads to monitor cardiac rhythm. Aliya system is monopolar and requires a grounding pad attached to the patient. When the system is activated by pressing a foot pedal, one treatment consisting of 100 pulses or packets of energy synchronized to the R-wave on electrocardiogram (EKG) is delivered over 5 minutes.

Aliya INUMI Flex needle was inserted through the robotic bronchoscope catheter into the target lesion. The needle was extended out of the protective sheath and adjusted to reach the distal border of the lesion using 3D imaging. The PEF treatment was then applied. The needle was withdrawn back 0.5 cm at a time, using 2D imaging, until the most proximal part of the lesion was treated. For a large lesion and if technically feasible, the needle was completely withdrawn and re-inserted into the target lesion using a different vector with a goal of maximal tumor volume ablation. For a different vector placement, tool-in-lesion confirmation was obtained using 3D imaging. The same steps were performed until the maximal tumor volume was treated.

The total procedure time was calculated from the bronchoscope insertion to removal time, the anesthesia time was from the induction to extubation, and PEF ablation time was the time utilized for ablation.

Lesion Response

The size of the treated tumor was assessed using the radiology reports or measured by electronic calipers on the CT chest. The volume of tumor was calculated using three dimensions. The *percentage of tumor volume ablated* was estimated based on the number of PEF treatments applied and INUMI Flex Needle ablation zone volume compared to the total tumor volume. The location of the lesion was determined by drawing a line through the lesion from pleura to mediastinum, and assigned in central, middle or outer

third of the lung, depending on where the center of the lesion was located on the line. The change in the size of the tumor and 18F-fluorodeoxyglucose (FDG) uptake on positron emission tomography (PET)-CT scan was assessed on the 3-month and subsequent follow-ups. Response Evaluation Criterion in Solid Tumors (RECIST) version 1.1 was used to evaluate the overall treatment response.²¹

Statistical analysis

Continuous variables were summarized using median and interquartile range (IQR) or range (minimum-maximum). Categorical variables were summarized using percentage.

Results

Six patients underwent bronchoscopic PEF of seven lesions (**Table 1**). Median age (interquartile range, IQR) was 75 (70-77.5) years, and half of the patients were females. There was one (16.7%) former and two (33.3%) current smokers. Three (50%) patients had NSCLC, one (16.7%) rectal, one (16.7%) thyroid, and one (16.7%) had urothelial cancer. The patient with thyroid cancer had two lung lesions. Two (33.3%) patients with NSCLC had recurrent stage I disease, while other patients (66.6%) had stage IV disease. All patients had progressed on standard oncologic therapies. Five patients (83.3%) had received previous radiation while the patient with thyroid cancer had two lesions and was treated with radioactive iodine previously but no radiation. Previously, one (16.7%) patient had received chemo-immunotherapy, one (16.7%) had chemotherapy, one (16.7%) had tyrosine kinase inhibitor, and one (16.7%) had radioactive iodine. One (16.7%) patient received concurrent and adjuvant chemo-immunotherapy. Two (33.3%) other patients received adjuvant chemo-immunotherapy on disease progression, three months after PEF. After three months of PEF ablation, two (33.3%) patients received chemotherapy and one was treated with immunotherapy (16.7%).

The median tumor size was 2.1 cm (range, 1.1-4 cm) as shown in **Table 2**. Six (85.7%) lesions were located in the central-third and one (14.2%) in the middle-third of the lung. All patients had diagnostic biopsies confirming malignancy on rapid on-site evaluation (ROSE) in the same procedure before the

PEF ablation was performed. Median (IQR) number of initial PEF treatments per tumor were 3 (2-3.5) and median number of treatment vectors were 1 (1-1.5). The median total procedure time was 68 (63.5-80) minutes, median anesthesia time was 82 (79.5-101) minutes, and median PEF ablation time was 15 (10-17.5) minutes. Median tumor volume ablation percentage was 100 (80.1-100). Some lesions did not achieve 100% ablation because of large lesion volume and technical inability to insert the PEF needle in a different optimal vector. One patient with tumor progression after PEF underwent a second PEF ablation. However, there was continued tumor progression on follow up PET-CT.

The patients were followed up with PET-CT scans for a median of 9.7 (5.4-10.1) months. The PET-CT scans were done every three months in all patients. On the final follow-up, four (57.1%) lesions increased in size, two (28.5%) decreased, and one (14.3%) resolved (**Table 3, Figures 1 and 2**). The median increase in the largest diameter for the lesions that grew was 0.7 (0.5-1) cm and the median decrease in the largest diameter of the lesions that shrunk was 1.1 (0.5-1.2) cm. Both the lesions increased in size in the patient with thyroid cancer. The decreased size or resolution of lesions was seen in the three patients with NSCLC. Only one of these patients received concurrent and adjuvant chemo-immunotherapy. All other patients received prior SBRT or radioactive iodine for thyroid cancer, one patient received prior chemotherapy and one patient had prior chemo-immunotherapy. On the last follow-up PET scan, FDG uptake increased for five (71.4%) lesions, decreased for one (14.3%), and resolved for one (14.3%). CT and PET scan changes were concordant in all patients, except one patient with NSCLC, where nodule decreased in size but FDG uptake on PET scan increased. Later clinical follow up after the end of the study period showed continued increase in the size and FDG uptake on PET-CT. Based on the RECIST 1.1 criteria at the final follow up, four (66.6%) patients progressed, one (16.7%) remained stable, and one (16.7%) had complete response. No abscopal effect was seen in the only patient with multifocal disease where the index lesion was treated.

No direct PEF related complications were observed within or beyond 30 days. There were no significant cardiac arrhythmias, bleeding, or pneumothorax. All the patients were alive at the time of final follow-up.

One patient was reintubated after the procedure for prolonged sedation but extubated after six hours, discharged within 24 hours, and continued to do well afterwards.

Discussion:

This is an early report of long-term safety and local control efficacy of bronchoscopic PEF ablation used in a clinical setting for patients with progressive lung cancer and cancers metastasized to the lungs. At a median follow-up of about 9 months, PEF ablation led to mixed local control but was safe with no direct complications.

The cohort of PEF ablation patients was particularly challenging, as they had tumor progression on standard of care therapy, and had limited treatment options as determined by a multidisciplinary team. All the patients, except one, had received radiation with recurrence in the prior field and further radiation was ruled out by the treating physicians. In this context for local control with PEF, four (57.1%) lesions increased in size, two (28.5%) decreased, and one (14.3%) resolved. The favorable response was seen only in patients with NSCLC, where treated lesions decreased in size or FDG uptake, while all of the metastatic lesions increased in size or FDG avidity. The NSCLC lesion that resolved was the smallest in the cohort, with the largest diameter of 1.1 cm. Lesion size may be an important predictor of response, with smaller lesions responding better to PEF ablation, and this should be evaluated in larger studies. The FDG avidity and tumor size changes were concordant on PET-CT scans, except in one patient with NSCLC, where tumor size shrunk but FDG uptake increased. This patient showed continued tumor progression afterwards. Therefore, increased FDG uptake after 3 months of PEF ablation may be a reasonable and sensitive marker for tumor persistence or progression.

A number of small studies have been recently published about the safety of bronchoscopic PEF, but ours is the first one to report local control.¹⁷⁻¹⁹ A prospective study evaluated the safety of PEF ablation using robotic bronchoscopy or percutaneous approach of 44 lung lesions in 30 treatment-naïve patients before the standard of care oncologic therapy.¹⁹ Over a follow-up of 30 days, one pneumothorax was

encountered with a percutaneous procedure. Unrelated to PEF, there was one case each of transient ischemic attack, respiratory failure related to anesthesia, and cervical spondylosis with radiculopathy. In another prospective, multicenter study evaluating the safety of PEF ablation, 36 patients with NSCLC underwent bronchoscopic or percutaneous ablation about 20 days before the surgical resection.¹⁷ These patients were compared with 8 patients who only had biopsy before the resection. Adverse events included two pneumothorax, one hemoptysis, and one patient with chest pain. Histopathological assessment of the resected tumor showed a cellular depletion zone with decreased or absent tumor cellularity surrounded by a transition zone with decreased cellularity compared to the untreated part of the tumor. Arterioles and bronchioles in the treatment zone were intact, as well as pleura when treatment was performed close to pleural surface. In addition, aggregates of CD20-positive B cells—tertiary lymphoid structures, were identified in and around the treatment zones, which were considered to promote local immune response against cancer cells.

There is more data on the safety as well as efficacy of the percutaneous PEF. In a multicenter, retrospective study, 41 patients with advanced lung cancer, progressive on standard therapy, were treated with percutaneous PEF ablation and the outcomes were compared with a propensity-matched cohort.²² Four patients had pneumothorax requiring chest tube and one had hemoptysis. The PEF group 1-year progression-free and overall survival were 63.2% and 74.3%, compared to 11.8% and 33% for the control group. Another retrospective study described the percutaneous PEF experience for 30 progressive tumors in 17 patients; two had NSCLC and nine had metastatic tumors in lungs.²³ Fifteen (50%) of the 30 treated lesions remained stable or decreased in size at a median follow-up of 3 months. Both treated NSCLC lesions remained stable. Twelve (50%) of the 24 non-treated lesions remained stable or decreased as well. Concurrent systemic chemo-immunotherapy was used in 14 (82%) of the 17 patients. There were 5 cases of pneumothorax, 3 pleural effusions, and one perihepatic hematoma which did not require drainage. One hemothorax required chest tube placement. In another retrospective study, 11 patients with 13 different metastatic lesions were treated with percutaneous PEF.²⁴ Three patients had metastases to lungs but no

participant had primary lung cancer. Residual disease and local or distant progression was seen in 10 out of 11 patients, and no abscopal effect was seen. One patient developed gastric artery pseudoaneurysm and hemoperitoneum, requiring vascular embolization. Another patient developed brachial plexopathy attributed to pressure due to body positioning for the procedure. Safety and feasibility of percutaneous PEF in tumors of multiple organs was assessed in 155 patients, including 77 with lung cancer.²⁵

Pneumothorax was seen in 21.5% patients with lung treatment, and 11.3% required a chest tube. One patient developed self-limiting hemoptysis, and another patient with pre-existing hemoptysis died of massive hemoptysis within 48 hours of the procedure. Our study adds to the current literature by reporting both the safety and local control efficacy of bronchoscopic PEF ablation. Not only was the bronchoscopic PEF safe in patients with progressive tumors located in central- and middle-thirds of the lung, but patients with NSCLC showed modest local control.

PEF ablation is suggested to improve immune response against the tumor. Immune synergy of PEF with chemo-immunotherapy in murine cancer model has been shown where PEF along with cisplatin and checkpoint inhibitor (α PD-1) increased the CD4⁺, cytotoxic CD8⁺, and CD4⁺ and CD8⁺ central memory T cells, slowed tumor growth, and improved survival.¹⁶ Another murine model study showed that PEF and immunotherapy, compared to radiofrequency ablation and immunotherapy, enhanced recruitment of M1 macrophages, natural killer cells, and dendritic cells and decreased regulatory T cells, leading to slowed tumor growth and better survival.¹⁵ Our study did not assess the immune markers and we did not see abscopal effect in the only patient with multifocal disease.

The strengths of our study include the first report of safety and local control efficacy of bronchoscopic PEF ablation in patients with progressive malignancies in the central and middle-thirds of the lung. We showed that PEF is safe but has mixed efficacy, with a potential benefit in patients with NSCLC. Our findings are similar to the study by Moore and colleagues, in which patients with progressive NSCLC who were treated with percutaneous PEF showed improved survival compared to propensity-matched controls.²² Kim and colleagues also showed stability of patients with progressive NSCLC treated with

percutaneous PEF.²³ We speculate that the potential efficacy in lung cancer seen in our cohort might be related to unique immunological characteristics of lung cancer, which has been shown to be one of the most responsive cancers to checkpoint inhibitors.^{3,26} The immune-responsiveness of lung cancer, synergized by PEF mechanisms described above, may be responsible for selective effectiveness of PEF in lung cancer. Limitations of our study include small sample size and a heterogenous cohort with no control group. Since most of our patients did not receive concurrent or immediate adjuvant systemic chemotherapy, we were unable to assess correlation between immunotherapy and PEF response. A randomized controlled trial is warranted to assess the role of PEF ablation in lung cancer and metastatic cancer to the lung, with immunologic markers and synergy evaluation with systemic immunotherapy.

Conclusion

Bronchoscopic PEF ablation is safe but has mixed local control efficacy. Future studies are needed to assess the impact of PEF on outcomes of patients with lung and metastatic cancer to the lungs.

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Table 1. Demographics of patients.

Patient demographics	Results (N=6)
Age, years- median (IQR)	75 (70-77.5)
Gender, female- n (%)	3 (50)
Race, white- n (%)	6 (100)
Smoker- n (%)	
Never	3 (50)
Former	1 (16.7)
Current	2 (33.3)
Cancer Type- n (%)	
Non-small cell lung cancer	3 (50)
Rectal	1 (16.7)
Thyroid	1 (16.7)
Urothelial	1 (16.7)
Stage	
Stage I	2 (33.3)
Stage IV	4 (66.7)
Prior oncologic treatment	
Chemotherapy	1 (16.7)
Chemo-immunotherapy	1 (16.7)
Tyrosine kinase inhibitor	1 (16.7)
Radiation	5 (83.3)
Radioactive iodine	1 (16.7)
Concurrent oncologic treatment*	
Chemo-immunotherapy	1 (16.7)
Adjuvant oncologic treatment	
Chemo-immunotherapy	3 (50)

* 4 weeks prior to 3 months post-PEF ablation

Table 2. Tumor characteristics and initial PEF ablation details

Tumor characteristics and PEF ablation	Results (Lesions, N=7)
Largest tumor diameter, cm- median (range)	2.1 (1.1-4.0)
Tumor volume, cm ³ - median (IQR)	2.5 (1.6-3.9)
Location- n (%)	
Central- third	6 (85.7)
Middle-third	1 (14.2)
Number of PEF ablation treatments per tumor- median (IQR)	3 (2-3.5)
Vectors of PEF ablation- median (IQR)	1 (1-1.5)
Percentage of tumor volume ablated, median (IQR)	100 (80.1-100)

Table 3. Tumor Response to PEF ablation

Tumor Response	Results (Lesions, N=7)
Interval between initial PEF ablation and follow-up, months- median (IQR)	9.7 (5.4-10.1)
Tumor size change on last follow-up compared to baseline- n (%)	
Increased	4 (57.1)
Decreased	2 (28.5)
Resolved	1 (14.3)
FDG-uptake change on last follow up compared to baseline-n (%)	
Increased	5 (71.4)
Decreased	1 (14.3)
Resolved	1 (14.3)
RECIST 1.1 criteria on the final follow-up-n (%)	N=6 patients
Progressive disease	4 (66.6)
Stable disease	1 (16.7)
Complete response	1 (16.7)

Figure Legends:

Figure 1: Local response of lung tumors treated with bronchoscopic PEF, including the change in size and FDG uptake on CT/PET scans at follow-up.

Figure 2: A and B- CT chest and PET scan of left upper lobe mass before treatment with bronchoscopic PEF. C and D- CT chest and PET scans of left upper lobe mass 6-months after treatment with bronchoscopic PEF.

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